POCKETSMART AI

Easy Project Documentation

Project Title: PocketSmart AI – Smart Budget & Recommendation Assistant

Technology: Python, FastAPI, HTML, CSS, JavaScript, Gemini AI

Purpose: To provide budget-based recommendations for home planning, parties and jewelry.

1. Introduction

PocketSmart AI is a simple AI-based recommendation system. It helps users plan their spending according to a given budget. The user enters a budget and preferences, and the system generates suitable suggestions.

The project mainly supports three planners:

- Home Interior Planner – recommends furniture and decoration items.
- Party Planner – suggests food, venue and decoration options.
- Jewelry Planner – suggests jewelry based on occasion, style and optional outfit image.

The original project documentation describes the system as a GenAI-powered platform using Gemini for personalized, budget-aware recommendations. ■filecite■turn0file0■L2-L14■

2. Objectives

- Help users manage a limited budget.
- Generate personalized recommendations using AI.
- Provide different planning modules in one application.
- Reduce the time required to search for suitable products or services.
- Give recommendations based on budget, preferences and occasion.
- Provide a simple and user-friendly interface.

3. Main Modules

Home Planner: User enters room type, budget and required items. The AI provides suitable home-interior suggestions.

Party Planner: User enters event type, budget and number of guests. The system suggests food, venue and decoration options.

Jewelry Planner: User enters budget, occasion and style. An outfit image can optionally be used for matching suggestions.

4. System Requirements

Software Requirements

- Python 3.x
- FastAPI
- HTML, CSS and JavaScript
- Gemini API
- VS Code or any Python IDE
- Web browser

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Keyboard and mouse

5. Basic Architecture

User → Frontend → FastAPI Backend → Gemini AI → Recommendation Result

The frontend collects user information. FastAPI handles requests and communicates with the AI layer. Gemini processes the budget and preferences and returns recommendations. The result is then displayed to the user.

6. Project Workflow

- Open the PocketSmart AI website.
- Register or log in to the application.
- Select Home, Party or Jewelry planner.
- Enter the required budget and preferences.
- Submit the form.
- Backend sends the information to the AI service.
- AI generates suitable recommendations.
- Recommendations are displayed to the user.
- Previous recommendations can be viewed in history.

7. Important API Routes

- /generate-home – Home recommendations
- /generate-party – Party recommendations
- /generate-jewelry – Jewelry recommendations
- /register – New user registration
- /login – User login
- /logout – Logout
- /history – Previous recommendation history

8. Simple Database / Data Flow

User details and session information are used to personalize the recommendation experience. The application can maintain recommendation history so users can review earlier queries.

9. User Interface Pages

- Home Page
- Register Page
- Login Page
- User Dashboard
- Home Interior Planner
- Party Planner
- Jewelry Planner
- Recommendation Pages
- Recommendation History

The source documentation describes these pages as part of the frontend and user experience. ■filecite■turn0file0■L247-L280■

10. Testing

Testing is performed using different budgets and user inputs for all three planners.

- Check whether the entered budget is accepted correctly.
- Check whether the correct planner produces results.
- Check whether empty or invalid inputs are handled.
- Check whether recommendations are displayed correctly.
- Test optional image input in the Jewelry Planner.
- Check the login, logout and registration functions.

11. Advantages

- Easy budget planning
- Personalized AI suggestions
- Multiple planners in one application
- Saves time
- Simple web-based interface
- Can support text and image inputs

12. Future Enhancements

- Connect to live product and service APIs.
- Add more categories such as travel and grocery planning.
- Improve recommendation accuracy.
- Add price comparison between platforms.
- Add mobile-friendly improvements.
- Add more advanced user preference tracking.

13. Conclusion

PocketSmart AI is a practical AI-based budget recommendation project. It combines a web interface, FastAPI backend and Gemini AI to provide recommendations for home interiors, party planning and jewelry selection. The project demonstrates how generative AI can be used to make everyday planning easier and more personalized.

The original documentation describes the same three major recommendation areas and the use of FastAPI with Gemini for budget-aware recommendations. ■filecite■turn0file0■L286-L303■